STAFF REVIEW · H446 206 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. A variable storing a running total.

2. Two keys map to the same initial bucket.

3. A reference to the first node.

4. A stack: the most recent operation must be undone first.

I Do — source worked steps

1. Predict — The last visited page is the first one that must be removed.

2. Explain — Use a stack: push visits and pop the most recent page.

We Do — guided CHECK

1. Explain
Use a queue, because FIFO removes the oldest item.

You Do — source mark schemes

1. Queue (FIFO).

2. Hash table: hashing IDs supports expected fast exact-key retrieval; stacks expose only the latest item.

3. Balanced BST allows ordered traversal and O(log n) search; a hash table has expected O(1) lookup but does not inherently keep keys ordered.

Exit Ticket — source answer

1. False. Collisions can make lookup much slower, sometimes O(n).